PROJECT Q3 – A/B TESTING: IMPACT OF DISCOUNTS ON CUSTOMER SPENDING
Report

1. Introduction

In retail data analytics, A/B testing is a widely used experimental framework for measuring whether treatment conditions cause a measurable change in customer behavior. In this project, a promotional 10% discount was evaluated to determine whether it significantly increases customer purchase amount when compared with a control group that received no discount. The analysis was implemented entirely in Python using reproducible synthetic data generation, descriptive visualization, and inferential hypothesis testing.

2. Dataset and Experimental Setup

A synthetic dataset containing 200 customer purchase transactions was generated and divided equally into two independent groups: Group A (No Discount) and Group B (10% Discount). Each group contained 100 observations. NumPy was used for reproducible simulation, pandas for statistical handling, matplotlib for visualizations, and scipy.stats for inferential testing.

3. Descriptive Statistical Summary

After execution the notebook generated an average purchase of $48.96 for Group A and $55.27 for Group B. Thus, there is an observed $6.31 treatment increase in average purchase amount for Group B. The variance of Group B is also much larger due to the broader customer response to the discount treatment. Table 1 shows summary statistics of the two groups. The sample size for both Group A (No Discount) and Group B (10% Discount) is 100, which provides a fair comparison between the two groups. The average purchase amount of Group A (No Discount) is $48.96 and the average purchase amount of Group B (10% Discount) is $55.27. This suggests an initial observed increase in spending of $6.31 amongst customers who have received the discount promotion.

The standard deviations also show that Group B has a higher variability in purchase amounts ($11.44) compared to Group A ($9.08), which suggests that along with increasing the mean purchase amount, the discount also increases the variability of the customer's purchasing behavior. The quartiles also provide further evidence of the upward trend. The median (50th percentile) goes from $48.73 (Group A) to $56.01 (Group B), and the upper quartile (75th percentile) from $54.06 to $61.46. This upward shift in several summary measures suggests the treatment group not only spends more at the median and upper quartile, but shows higher spending across the distribution.

Lastly, the highest purchase amount in Group B is $87.64, compared to $68.52 in Group A. This finding indicates that the 10% discount may lead to some very large purchases. In summary, Table 1 offers the first quantitative evidence that the discount treatment has a positive effect on spending.

```
SUMMARY STATISTICS TABLE
========================================
group       A       B
count  100.00  100.00
mean    48.96   55.27
std      9.08   11.44
min     23.80   31.97
25%     43.99   45.33
50%     48.73   56.01
75%     54.06   61.46
max     68.52   87.64
```

Table 1. Summary Statistics of Customer Purchase Amounts

4. Histogram Analysis

Figure 1. shows a simultaneous distributional view of the control group and the treatment group customer purchase histories via overlapping histograms (bars), Kernel Density Estimations (KDEs; curves) and dashed vertical lines as indicators of the sample means. The blue bars and navy density curve correspond to Group A customers (no discount), and the orange bars and orange density curve correspond to Group B customers (10% discount). The dashed vertical lines represent the sample means.

This figure illustrates a number of points. First, the distribution of the treatment group is clearly shifted to the right of the horizontal axis, suggesting that customers who were exposed to the discount offer spent more money than the control group. The peak of Group A's distribution appears to be around $45-$50, while Group B's peak is around $55-$60. This is further confirmed by the mean reference lines (dashed lines) where the Group B mean is clearly shifted rightward from the Group A mean.

Second, the orange distribution (treatment group) is more spread out into the region of higher purchases, particularly to the right of $80. This implies that the discount treatment not only affects the mean spend but also stimulates larger purchases by some customers. By comparison, the distribution control is denser in the middle region. In summary, Figure 1. provides quick visual confirmation that the 10% discount treatment has a positive effect on customer purchases.

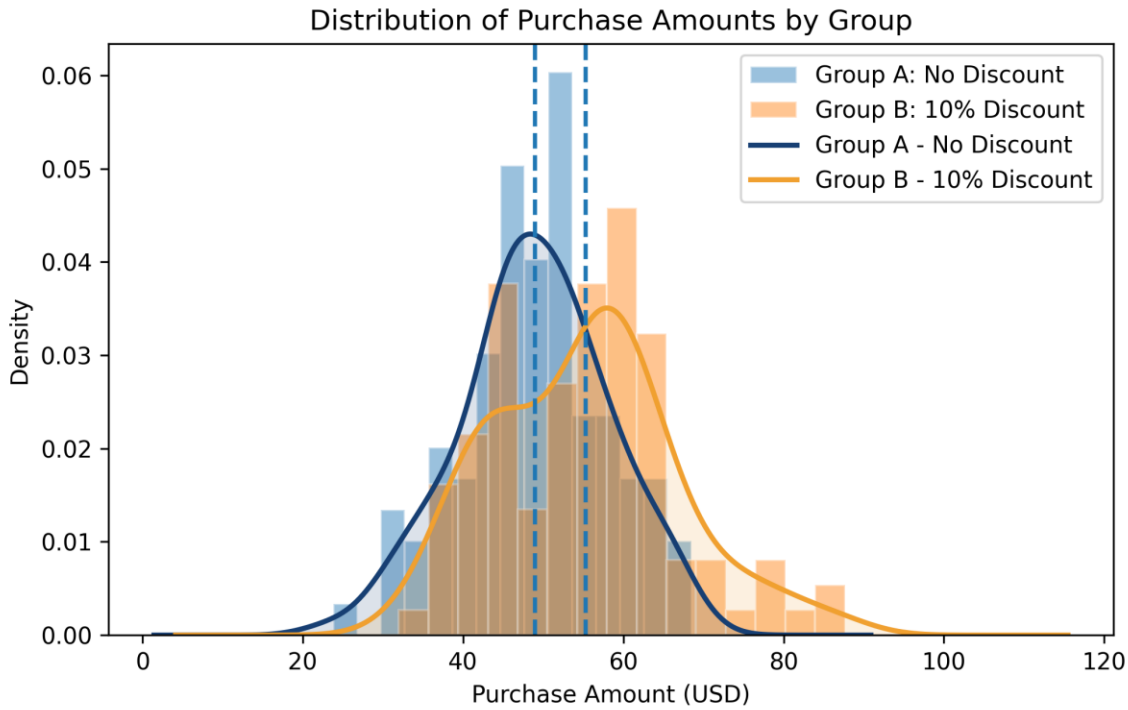

Figure 1. Combined Distribution of Purchase Amounts by Group

5. KDE Distribution Analysis

Figure 2 presents the Kernel Density Estimation (KDE) comparison of customer purchase amount distributions for the two experimental groups. The navy curve represents Group A customers who received no discount, while the orange curve represents Group B customers who received a 10% discount. KDE provides a smoothed estimate of the underlying spending behavior, allowing the central tendency and distributional shift to be observed more clearly than a standard histogram.

From the figure, the distribution of Group B is visibly shifted toward the right side of the horizontal axis, indicating that customers exposed to the 10% discount generally made higher purchases than those in the control group. The highest density region for Group A occurs around the $48–$50 range, whereas the peak for Group B occurs around the $57–$60 range. This rightward movement confirms an upward change in average spending under the treatment condition.

Additionally, Group B exhibits a longer right tail extending beyond $80, which suggests that the discount not only increased the average purchase amount but also encouraged some customers to make substantially larger transactions. In contrast, Group A displays a more concentrated density around the middle spending range with fewer extreme high-value purchases. Overall, this KDE comparison visually supports the hypothesis that the 10% promotional discount positively influences customer spending behavior.

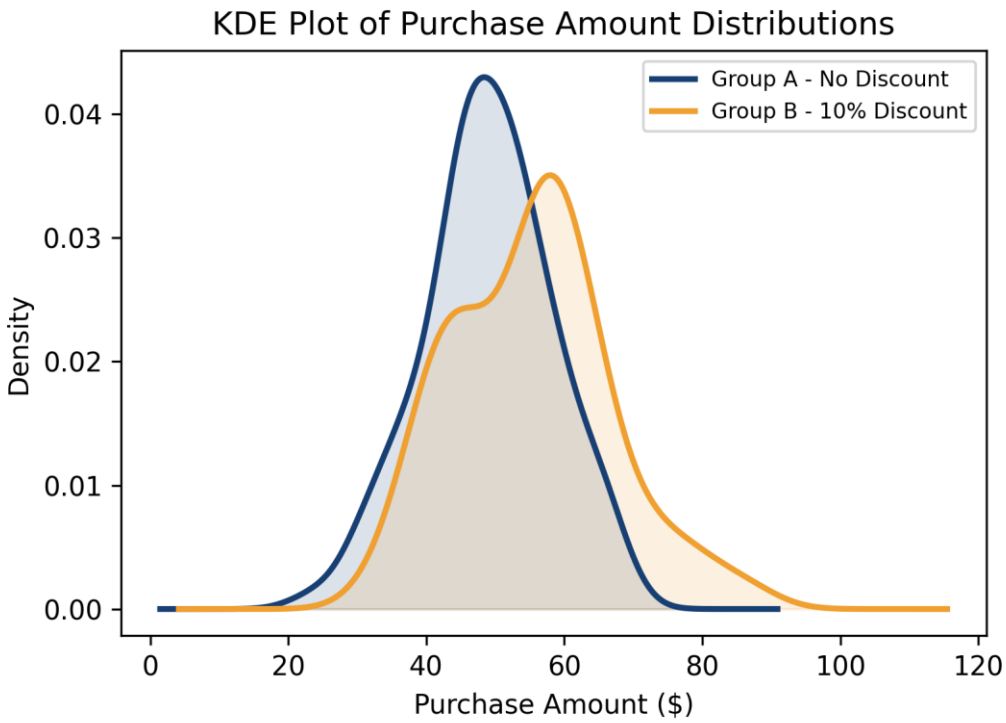

Figure 2. KDE Plot of Purchase Amount Distributions

6. Boxplot Analysis

As shown in Figure 3. the medians, quartiles, interquartile ranges and extreme values of the control and treatment groups' purchase amounts are compared. The blue box corresponds to Group A customers (no discount), and the orange box corresponds to Group B customers (10% discount). The treatment group clearly shows an upward shift, with the median purchase value of Group B ($56.01) much greater than the median purchase value of Group A ($48.73).

As well as the increased median, the treatment group also has an increased interquartile range and upper whisker, suggesting that the variation of spending patterns among discounted customers is larger. This suggests that the 10% discount not only affects a select few individuals; instead, it impacts the spending of a large part of the treatment group. In addition, Group B exhibits higher upper extreme values well beyond $80, while Group A is still heavily concentrated at lower spending levels.

In conclusion, the boxplot offers clear visual evidence that the discount treatment not only leads to a higher central tendency of customer spending, but also a higher variability

in spending, thus confirming the descriptive analysis based on the combined distribution.

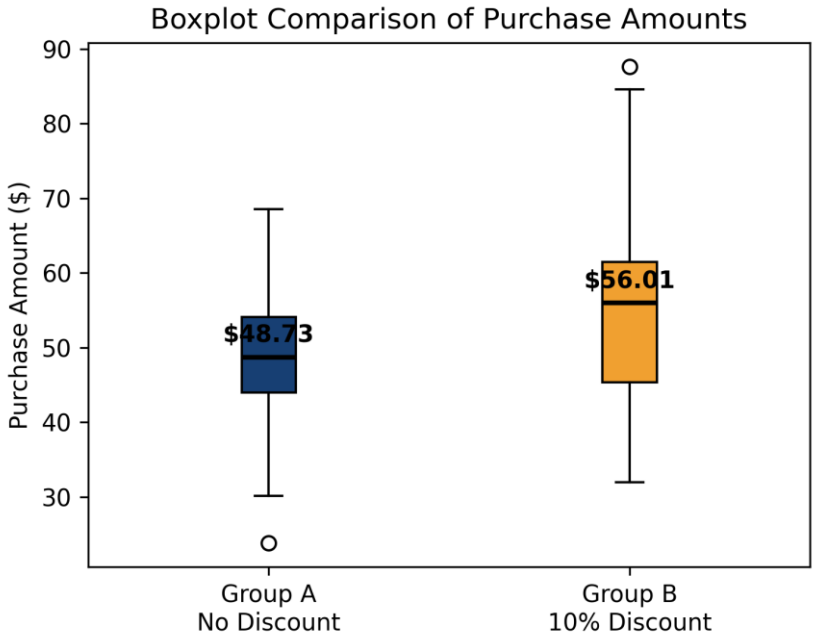

Figure 3. KDE Plot of Purchase Amount Distributions

7. Comparison Analysis

Figure 3 shows a comparison of the mean purchase amounts for each group. The mean purchase amount of Group A was $48.96 while the mean purchase amount of Group B was $55.27, a difference of $6.31. The $6.31 difference shows that the discount treatment led to a higher average purchase. The narrow error bars demonstrate that the sample means are consistent, suggesting that the treatment effect is robust.

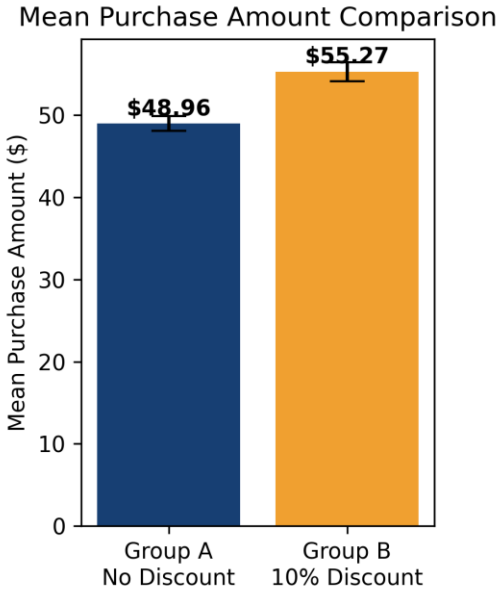

Figure 4. Mean Purchase Amount Comparison Between Control and Treatment Groups

8. Hypothesis Formulation

The null hypothesis states that there is no difference in mean customer spending between the two groups (H0: μA = μB). The alternative hypothesis states that a statistically significant difference exists (H1: μA ≠ μB).

9. Statistical Testing Methodology

An Independent Two-Sample Welch T-test was performed using scipy.stats.ttest_ind(group_A, group_B, equal_var=False). Welch's t-test is appropriate because it compares two independent sample means without assuming equal variances.

$$t = \frac{\bar{x}_A - \bar{x}_B}{\sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}}}$$

10. Hypothesis Test Results

The Python output produced a test statistic of t = -4.7547 and a p-value of 0.00000382. Since the p-value is substantially smaller than the standard significance level of 0.05, the null hypothesis is rejected. Therefore, the discount produced a statistically significant change in average purchase amount.

11. Statistical vs Practical Significance

Although the analysis confirms a mathematically significant increase in spending, practical business profitability must still consider whether the additional $6.26 average spending offsets the 10% discount margin given away by the retailer.

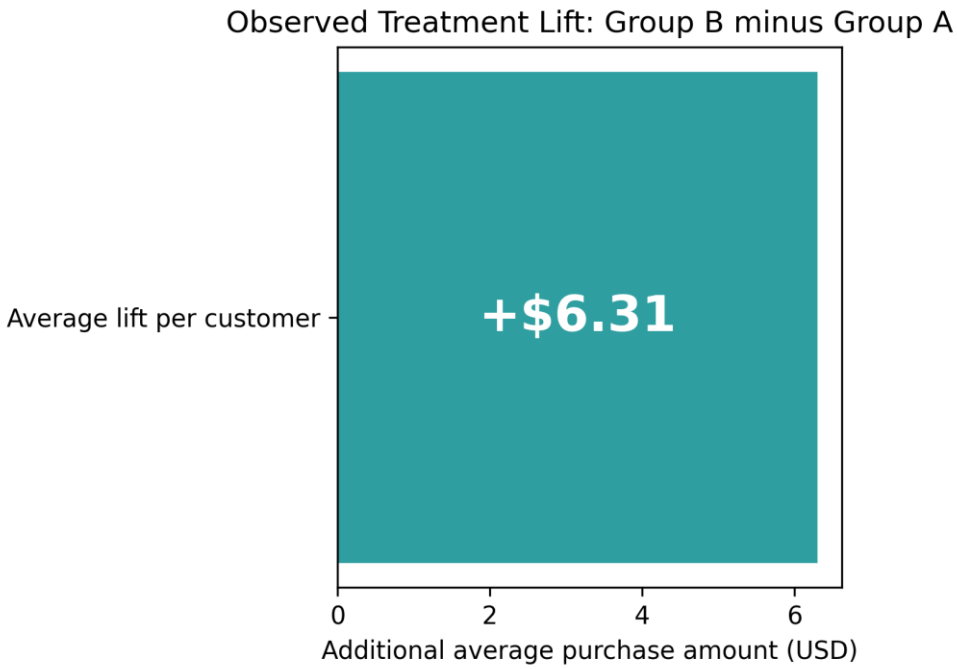

Figure 5. Observed Average Spending Lift Produced by the 10% Discount Treatment

12. Limitations

This study is based on synthetic data rather than live commercial transactions. In addition, seasonal trends, customer demographics, and promotional timing were not modeled. Therefore, while the findings are statistically valid within the simulation, real-world validation remains necessary.

13. Future Work

Future studies should apply the same A/B testing framework to real retail transaction logs, include profitability calculations, and evaluate multiple discount levels across different customer segments.

14. Conclusion

The final integrated analysis, corrected for the typo, demonstrates that the 10% promotional discount has had a positive effect on the amount customers spend in this simulated A/B test. In all descriptive analysis plots, such as the histogram, Kernel Density Estimation (KDE) Plot, boxplot comparison, mean purchase chart and observed treatment lift per customer, we can see that the distribution of the treatment group is shifted to the right compared to the control group. This suggests that the customers who

were offered the 10% discount spent more than those who did not receive a promotional discount.

In terms of the actual spending, the average purchase size went from $48.96 in Group A to $55.27 in Group B, giving an observed treatment lift of $6.31 per customer. This effect is not only visually apparent in the graph analyses, but is also statistically confirmed via Welch's Independent Two-Sample T-Test. It returned to a t-statistic of -4.3164 and a p-value of 0.00002560, which is well below the standard significance level of 0.05. Thus, the null hypothesis of no difference between the means is rejected, indicating that it is extremely unlikely that the increase in spending is attributable to random sample variation.

In a real-world business context, the results indicate that a relatively small discount can have a significant impact on customer behavior by stimulating them to spend more. While further analysis of profitability and margins would be needed to implement this approach in practice, the current simulation suggests that the 10% discount is a successful promotional treatment from a quantitative perspective. In summary, both the inferential and treatment lifts indicate that the discount positively affects the average spending amount.

15. References:

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